The "E" suffix denotes extra Gen 5 lanes for NVMe drives. For high-end Ryzen 9 or Ryzen 9000 CPUs choose X670E for maximum bandwidth; B650 suits mid-range builds.

Intel's LGA-1851 platform includes Z890, B860 and H810 chipsets. Z890 boards provide sixteen CPU lanes for graphics, four for NVMe and generous chipset lanes, plus CPU and memory overclocking. B860 retains most connectivity but omits CPU overclocking, making it ideal for mainstream gaming and productivity. H810 drastically reduces lanes and ports and lacks overclocking, making it suitable only for basic systems. When planning your build, list your GPU, NVMe drives and peripherals; choose the chipset that meets your expansion and performance goals.

Form factor choices

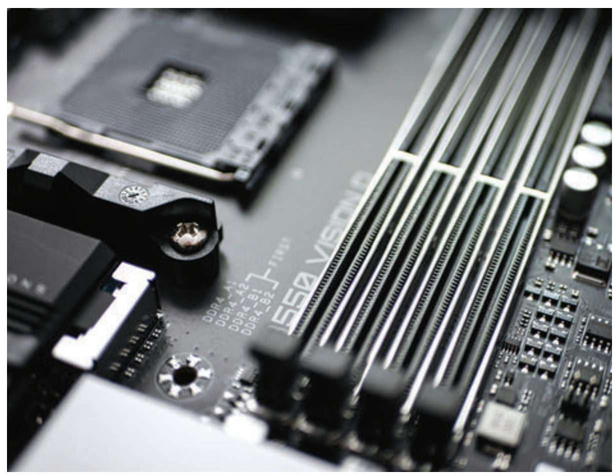
The form factor determines a board's size and layout. ATX boards are 305 × 244 mm with four DIMM slots, multiple PCIe x16 slots and ample connectors, making them ideal for high-end systems with multiple GPUs or extensive storage. They require mid-tower or full-tower cases and offer the broadest selection of models. Micro-ATX shrinks to 244 × 244 mm and typically provides four RAM slots but only two PCIe x16 slots and fewer SATA and USB ports. It fits into smaller cases, trading some expansion capacity for a lighter build. Mini-ITX is the smallest at 170 × 170 mm, usually offering two RAM slots, one PCIe x16 slot and limited M.2 and SATA ports; it supports very compact systems but leaves minimal room for upgrades. Choose the form factor that matches your case and the number of cards and drives you intend to install.

Power delivery and VRMs

Motherboards regulate voltage using a Voltage Regulator Module (VRM) composed of MOSFETs, chokes, capacitors and a controller chip. High-quality VRMs provide stable power, enabling CPUs to maintain boost clocks and overclock; weak VRMs can throttle the CPU and run hot. Boards advertise power phases such as 16+2; more phases spread the current across multiple components, supporting high-core-count processors. Top-tier boards like ASUS's Z890 range use dual 8-pin power connectors and VRMs with over 22 power stages cooled by large heatsinks. When pairing a high-TDP CPU like a Core i9 15900K or Ryzen 9 9950X, select a board with a robust VRM and adequate cooling; mid-range CPUs are fine with simpler VRMs.

PCI Express, M.2 and storage

PCI Express lanes connect your graphics card, storage and other peripherals. CPUs provide sixteen lanes for the GPU and four for a primary NVMe drive, while the chipset contributes additional lanes. X670 boards allocate more lanes than B650; similarly Z890 has more lanes than B860. Populate PCIe and M.2 slots carefully – using multiple NVMe drives may reduce the bandwidth of secondary PCIe slots, and some SATA ports may be disabled when certain M.2 slots are occupied.



An M.2 drive is a gum-stick-shaped module that bolts directly to the board. Common lengths range from 42 mm to 110 mm, with 2280 (22 × 80 mm) being the most popular. M.2 SATA drives use the chipset's SATA controller and max out around 550 MB/s. M.2 NVMe drives leverage PCIe lanes and achieve multi-gigabyte per second speeds – next-generation models can reach 14.5 GB/s. M.2 sockets are keyed: B-key slots support PCIe x2 or SATA; M-key slots offer PCIe x4/x8 bandwidth needed for high-performance NVMe drives; B+M-key drives fit either slot but are limited to x2 speeds. Consult your manual to ensure each drive operates at its full potential and note any lane sharing.

Front-panel and USB connectivity

The board's headers provide connectors for case buttons, LEDs and front-panel ports. Most boards use a nine-pin F_PANEL header for the power and reset switches and for power/HDD